PALRAM



PALSUN® Technical Guide

Flat Solid Polycarbonate Sheet

Build On.

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PALSUN® Product Range

Some of the products' features below can be combined. Please contact your Palram dealer for further information.

Product	Description	Features & Applications
PALTUF®	UV stabilized, general purpose, flat solid polycarbonate sheet.	Recommended for indoor use only.
PALSUN®	Flat solid polycarbonate sheet with UV protective layer on one side.	Suitable for both exterior and indoor applications.
PALSUN® UV2	Flat solid polycarbonate sheet with UV protective layer on both sides.	Recommended for applications that involve UV exposure on both sides (e.g. exterior light boxes)
PALSUN® Embossed	Flat solid polycarbonate sheet with optional surface textures: Embossed (E-102), Prismatic, Hair-cell, Matte - Anti-glare effect.	Recommended for applications that involve UV exposure on both sides (e.g. exterior light boxes)
PALSUN® FR	Sheet with higher fire resistance rating (e.g. UL 94 V-0).	Recommended for applications in populated areas.
PALSUN® Solar Control	Sheet with integrated heat-blocking layer. This metallic reflective layer that transmits less Infrared radiation and reduces heat buildup.	Available with 20, 35, or 50% light transmission.
PALSUN® Breeze	This advanced heat-blocking SolarSmart™ tint offers a clear view due to its high clarity.	Can be tailored to achieve different light and heat transmissions.
PALGARD™	Sheet with abrasion resistant layer on one or both sides.	Recommended for applications in high traffic areas, harsh chemical environments or those requiring anti-vandal properties.
PALSHIELD™	Containment and bullet resistant panel	Recommended for security glazing in banks and government facilities, public institutions, gas stations and convenience stores.

Notes:
1. All the above sheets are supplied with a protective polyethylene (PE) film on both sides (one side upon request), with the UV protected side clearly marked. This film should be removed immediately after installation.
2. For transportation, handling and storage instructions and recommendations, please refer to Handling and Storage Guidelines for Palram Sheets.
3. PALSUN sheets are backed by a 10 years limited warranty, available upon request.
4. Most PALSUN sheets are available in the transparent, translucent or opaque form, in a variety of colors, either standard or custom ordered.

Standard Dimensions*

		Surface Finish				
Thickness (mm)	**Width x Length** (mm)	**Smooth** Both sides	**Embossed** Both sides	**Matte** One side	**Hair Cell** One side	**Prismatic** One side
1	1220 x 2440 1250 x 2440	✔		✔		
1.5		✔	✔	✔		
2	1220 x 2440 1250 x 2440 2050 x 3050	✔	✔	✔		
2.5 - 6		✔	✔	✔	✔	✔
8		✔	✔		✔	
9 - 12.7		✔			✔	
2 - 18	2450 x 3050	✔				

*Other dimensions and specifications are available upon request, subject to a minimum order.

Colors

Color Group	Description	Colors
Clear	Transmits up to 90% natural daylight, resulting in high lighting within the structure.	Clear
Transparent	Low haze colors offering high clarity. Breeze and Smart are SolarSmart™ colors with that reduce heat buildup while allowing a clear view through the sheet (see page 9 for more details on SolarSmart products).	Solar Gray, Bronze, Blue, Green, Red, Smart Blue*, Smart Green*, Bluish Breeze*
Translucent	**White Opal:** Transmits 11-50% visible light with high light dispersion, produces mild and even lighting within the structure. **Diffuser:** Transmits 50% visible light with high light dispersion, produces diffused and consistent lighting within the structure. **LB (Light Box):** Diffuser sheet for illuminated signs and display applications, offering 44-50% light transmission.	Yellow, White Opal, White Diffuser, LB (Light Box Diffuser), Mint Green, Solar Control (Solar Metallic Grey), Solar Olympic, Solar Ice, Red
Opaque*	Colors that transmit very little to no light.	Dark Green, Red Brick, Black, Dark Blue, Cream Ral 9001, Light Grey Ral 7035, Dark Grey, Brown, Off-White

* Subject to minimum quantity. Custom colors and light transmissions are also available in this manner.
** Colors shown above are a reproduction of the actual color. To accurately represent the colors, contact your Palram distributor and request a sample color chip.

SolarSmart™ - Energy Efficiency

SolarSmart™ are energy-efficient colors break the traditional ratio between light transmission and shading coefficient. SolarSmart™ panels block Infrared energy that causes heat buildup, and transmit "cool light" that reduces air-conditioning and lighting costs.



Physical Properties

The following table displays physical properties of 3mm (0.12 inch) PALSUN and PALTUF sheets.

Property	Method**	Conditions (U.S. Customary)*	Units - SI (U.S. Customary)*	Value (U.S. Customary)*
Physical				
Density	D-792		g/cm³ (lb/ft³)	1.2 (75)
Water Absorption	D-570	24 hr. @ 23°C	%	0.15
Mechanical				
Tensile strength at yield	D-638	10 mm/min (0.4 in./min)	MPa (psi)	62.5 (9,100)
Tensile strength at break	D-638	10 mm/min (0.4 in./min)	MPa (psi)	65 (9,500)
Elongation at yield	D-638	10 mm/min (0.4 in./min)	%	6
Elongation at break	D-638	10 mm/min (0.4 in./min)	%	>80
Tensile Modulus of Elasticity	D-638	1 mm/min (0.4 in./min)	MPa (psi)	2,300 (290,000)
Flexural Modulus	D-790	1.3 mm/min (0.052 in./min)	MPa (psi)	2,350 (343,000)
Flexural Strength at Yield	D-790	1.3 mm/min (0.052 in./min)	MPa (psi)	93 (13,600)
Notched Impact Strength Izod	D-256	23°C (73°F)	J/m (ft·lbf/in.)	800 (15)
Notched Impact Strength Charpy	D-256	23°C (73°F)	J/m (ft·lbf/in.)	800 (15)
Impact Falling Weight	ISO-6603/1b		J (ft·lbf)	158 (117)
Rockwell Hardness	D-785		R scale / M scale	125 / 75
Thermal				
Long Term Service Temperature			°C (°F)	-50 to +100 (-175 to +212)
Short Term Service Temperature			°C (°F)	-50 to +120 (-175 to +250)
Heat Deflection Temperature	D-648	Load: 1.82 MPa (264 psi)	°C (°F)	135 (275)
Vicat Softening Temperature	D-1525	Load: 1 kg (2.2 lb)	°C (°F)	150 (300)
Coefficient of Linear Thermal Expansion	D-696		mm/m °C (Mil/in. °F)	0.065 (0.036)
Thermal Conductivity	C-177		W/m K (Btu·in/hr·ft²·°F)	0.21 (1.46)
Specific Heat Capacity	C-351		kJ/kg·°K (Btu/lb·°F)	1.26 (0.31)
Optical				
Haze	D-1003	Clear Sheet	%	<0.5
Light Transmission	D-1003	Clear Sheet	%	89
Refractive Index	D-542	Clear Sheet		1.586
Yellowness Index	D-1925	Clear Sheet		<1
Electrical				
Dielectric Constant	D-150	50 Hz		3.0
	D-150	1 MHz		2.9
Dissipation Factor	D-150	1 KHz		0.001
	D-150	1 MHz		0.01
Dielectric Strength Short Time	D-149	500 V/s	kV/mm (V/mil)	>30 (>770)
Surface Resistivity	D-257	Keithley	Ohm	10^{16}
Volume Resistance	D-257	Keithley	Ohm-cm	10^{17}

* Conditions, units and values in U.S. Customary units are presented in the table within parentheses.
** ASTM except where noted otherwise.

Impact Strength

PALSUN sheets are manufactured from polycarbonate, the most versatile, toughest transparent thermoplastic. PALSUN has 200 times the impact strength of glass, offering excellent protection against riots and public disturbances, breaking & entering or acts of vandalism.

PALSUN can endure attacks by rocks, clubs, hammers and thrown objects, and still to retain its original shape, maintain its integrity with minimal indentations to its surface.

The amount of damage depends on the object mass and energy, and sheet's thickness. PALSUN sheets will retain these energy-absorbing properties over a wide temperature range (50° to + 100° C).

Typical impact failure energy of PALSUN® sheets*

Thickness mm	Energy at Failure E_{50} (Jouls)	Type of Failure
2	110	100% ductile
3	150	100% ductile
4	190	100% ductile
5	290	100% ductile
6	400	100% ductile
8 - 18	N.B.**	N.B.**

* According to ISO 6603/1 1985(E): Determination of multi-axial impact behavior of rigid plastics.

Falling Dart Method

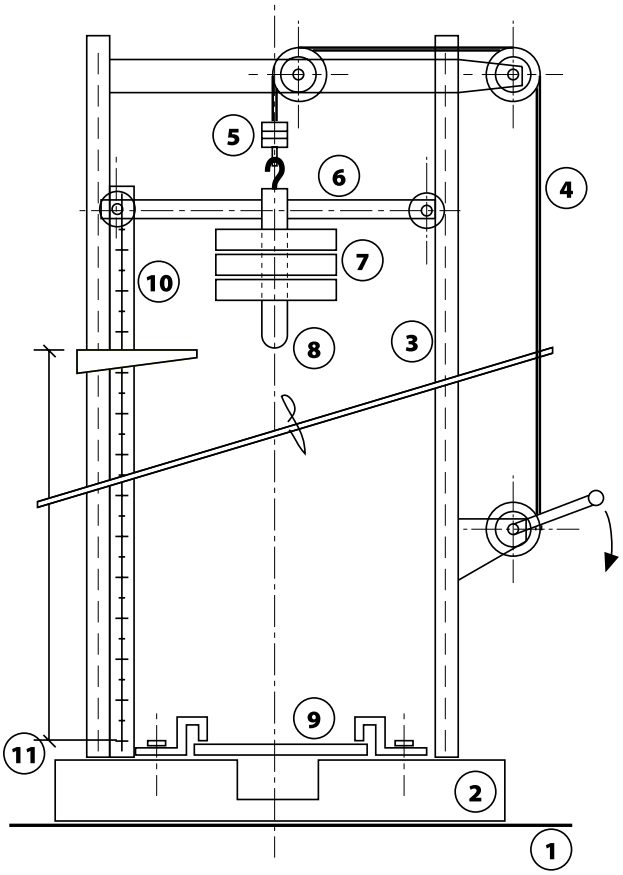
A uniform energy increment is employed during testing. Energy is decreased or increased by uniform increment after testing each specimen, depending upon the result (failed / not failed) observed for the former tested sample. A 20 mm diameter dart, weighing 8 kg, with a rounded tip, is raised to a certain height and released to fall on a suitably sized sample.

Principles: Impact strength is determined by the known weight and height. Adjustment is done by altering height while using a constant mass.

E50: 50% of Impact Failure Energy. The energy that will cause 50% of the tested samples to fail.

N.B.:** No Break. The energy required to break the sample is greater than what the test instrument can deliver.

Falling Dart Impact Testing Device (Schematic Figure)



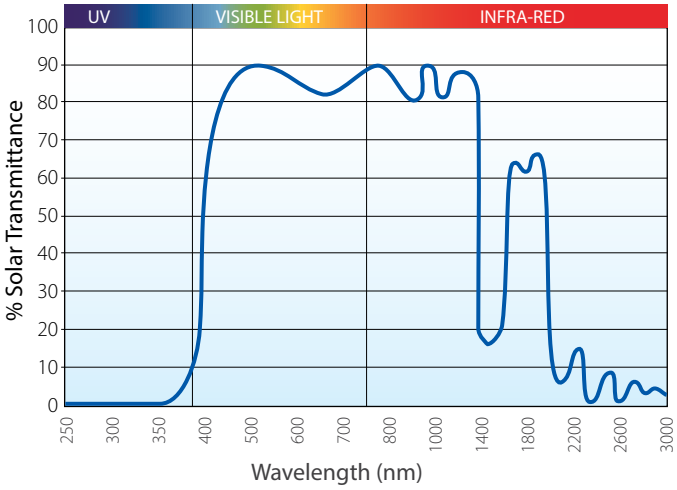
Legend

1. Leveled floor
2. Stablized base
3. Supporting guidrails
4. Lifting mechanism
5. Disengagement mechanism
6. Guidance bar
7. Changeable weight
8. 20 mm diameter head falling dart
9. Tested specimen
10. Calibration bar
11. Changeable drop height

Optical Characteristics

Ultraviolet (UV) Radiation Blocking - PALSUN sheets completely block out potentially harmful UV radiation and a significant portion of Infrared (IR) radiation. Over the visible light range, a typical 3 mm (0.125 inch) thick clear PALSUN sheet transmits about 89% (average) of incident light, as seen in the attached graph.

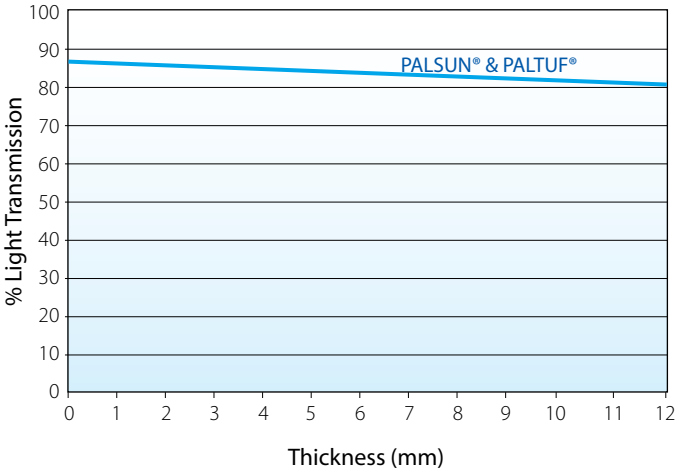
Figure 2:
Solar Transmittance of Clear 3mm PALSUN®



Light Transmission Versus Thickness

Light transmission decreases slightly with increased thickness (see attached graph).

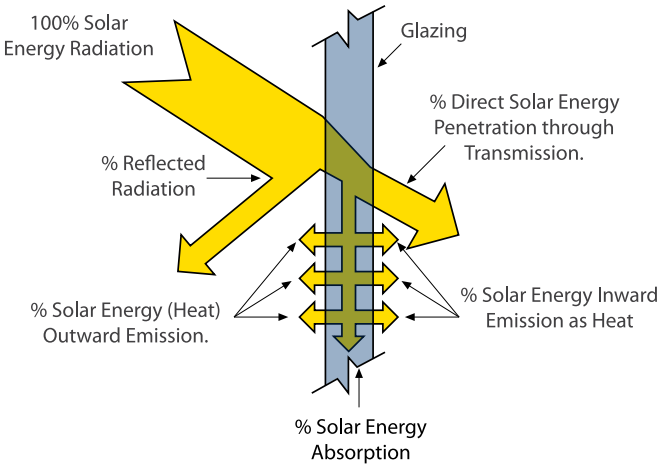
Figure 3: Light Transmission Versus Thickness



Solar Transmission Properties

Solar energy transmission is an extremely important consideration with transparent materials. Geographic location and typical thermal/optical properties of the specific glazing are the main factors influencing solar heat gain. Various types of PALSUN - textured, tinted, opal, diffused, and heat blocking SolarSmart™ sheets - can be used to deliver the exact quantity and quality of light desired. Each of these products transmit different amounts of direct light in varying levels of light diffusion, which may help to spread the light throughout the structure or enclosure. The sheets also vary in their selectivity index (SI) values, which determines how efficiently they keep heat out while letting more "cool light" in (See next page for more information on SolarSmart™ products). Although colors and tints reduce the percentage of visible light transmitted through the sheets, but solar energy is still absorbed by the glazing itself, and in turn transferred by convection and far IR radiation from the heated glazing into the building. PALSUN sheets with embossed or matte surfaces, or diffuser colors, diminish glare and dazzle, preventing damage by direct irradiance. However solar energy is still transmitted through and increases the solar heat gain inside the structure.

Figure 4: Solar Radiation Schematic Behavior Through Light Transmitting Material



Color*	% Light Transmission ASTM D-1003	%Haze ASTM D-1003	Solar Heat Gain (SHGC) ASTM E-424-71	Shading Coefficient ASTM E-424-71
Clear	90	<1	0.87	1.00
Bronze	20	<1	0.45	0.52
	35	<1	0.56	0.64
	50	<1	0.65	0.75
Solar Grey	20	<1	0.44	0.51
	35	<1	0.56	0.64
	50	<1	0.65	0.75
White Opal	28	100	0.32	0.37
White Diffuser	80	100	0.87	1.00
Solar Ice	20	100	0.37	0.45
Solar Control	20	67	0.33	0.36
	35	52	0.45	0.52
	50	50	0.54	0.61
Solar Olympic	20	35	0.41	0.47
	35	20	0.52	0.60
	50	63	0.63	0.73
Smart Green	70	42	0.58	0.67
Smart Blue	70	42	0.57	0.65
Bluish Breeze	70	42	0.55	0.63

*Values in the table above relate to 3mm Sheet. Further information on additional products is available upon request.

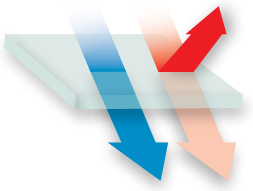
Terminology Used in the Table

Solar Radiation: The solar spectrum ranging from 300 nm to 2400 nm. Included are UV, visible and Near-IR radiation.
Visible Light Radiation: The portion of the light spectrum whose wavelength ranges from 400nm to 780nm.
% Light Transmission (ASTM D-1003): Percentage of incident visible light that passes through an object.
% Solar Heat Gain (SHGC): The percent of incident solar radiation transmitted by an object which includes the direct solar transmission plus the part of the solar absorption reradiated inward.
Shading Coefficient (ASTM E424-71): The ratio of the total solar radiation transmitted by a given material to that transmitted by normal glass, whose light transmission is 87%. It can be approximately calculated by: $\%ST + (0.27 \times \%SA) = \%ST_t$ $SC = (1.15 \times ST_t)/100$

SolarSmart™ Technology - Efficient Daylighting



SolarSmart™
Selective Solar Control Technology



Promoting Energetic Efficiency and Well Being

SolarSmart technology defies standard transmission of solar energy in transparent sheets and allows more versatile color and solar transmission specification per project. Unlike regular tints, SolarSmart sheets and panels admit more natural daylight while reflecting outwards Infrared radiation that creates heat. This characteristic breaks the traditional link between shading coefficient and light transmission, allowing a different perspective on the specification of natural light in architectural design.

SolarSmart tints allow better use of natural lighting without sacrificing the interiors. More natural light results in a healthier and more productive ambience. Energy saving is also promoted through reduction of both illumination and air conditioning requirements.



Color Specification

SolarSmart™ tints can be applied to any Palram transparent polycarbonate sheet or panel system, including PALSUN. The tints can be blended with any color to tailor the desired appearance and solar properties.

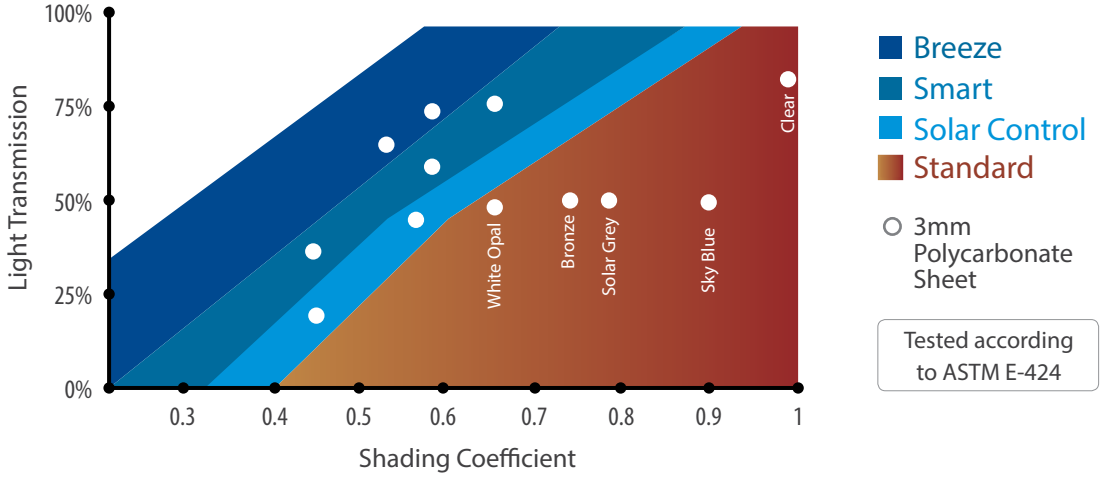


Technology Groups

The SolarSmart product range includes 3 technology groups, which have different characteristics and appearance. For more information please refer to the SolarSmart™ Technology brochure.

Efficiency Comparison

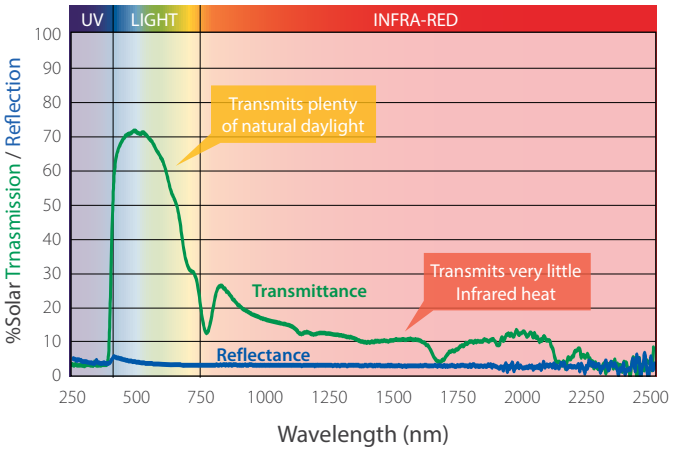
The graph below demonstrates the efficiency of SolarSmart products in comparison with clear and standard colored sheets. The graph shows how SolarSmart™ tints enable higher light transmission specification while maintaining or decreasing shading coefficient values.



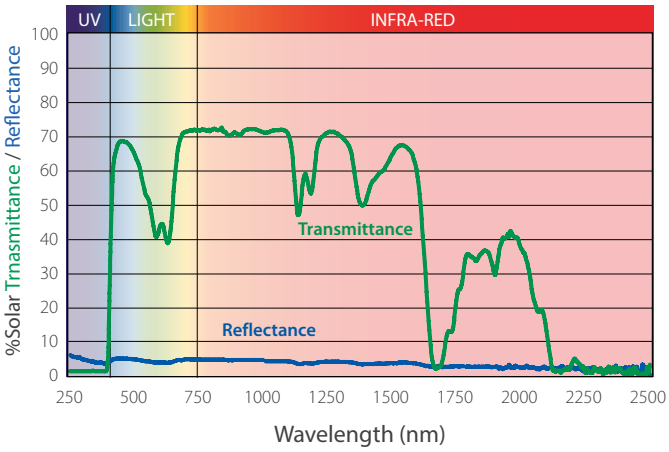
Heat-Blocking with High Clarity by PALSUN® Breeze

PALSUN glazing with Breeze tint transmits plenty of incident daylight (70%), while absorbing/blocking a large portion of the Infrared radiation, thus significantly reducing heat buildup within the structure. PALSUN Breeze also provides a clear view due to its high transparency, which is uncommon for heat-blocking products.

PALSUN® 3mm Breeze 70%LT
Solar Performance Graph (250-2,500nm)



PALSUN® 3mmTrans. Blue 68%LT
Solar Performance Graph (250-2,500nm)

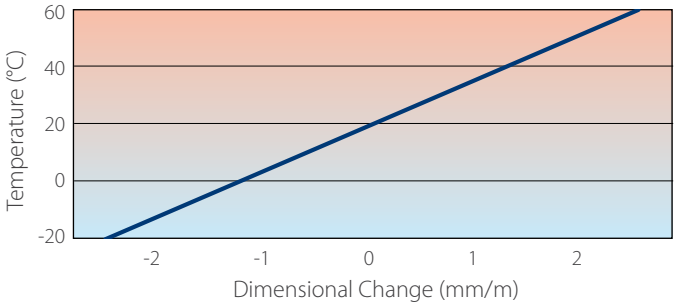


Thermal Characteristics

Thermal Expansion

The thermal expansion of PALSUN sheets is higher than that of glass. This important factor must be taken into account when mounting the sheets. The graph on the right shows the degree of expansion/contraction as a function of temperature.

Expansion/Contraction with Temperature



Service Temperature

The temperature range over which the characteristics of PALSUN are retained extends from -50°C to +120°C (-60°F to +250°F) for short periods and from -50°C to +100°C (-60°F to +210°F) for long periods. This range of temperatures make PALSUN sheets suitable for use in most climates.

Thermal Insulation

On very hot days, the surface temperature of the sheet might reach up to 60°C (140°F). The U-value characterizes the degree of thermal transmittance offered by a given glazing material, so higher U-values are associated with materials that are poor insulators and result in a greater loss of heat. The following table compares the U-values of glass and PALSUN sheets of equivalent thicknesses. Thicker sheets of a given material will offer greater thermal insulation and be characterized by a lower U-value and reduced heat loss. For any given thickness, the U-value of PALSUN sheet is lower than that of glass. This means that heat loss from the building interior, as well as penetration of heat or cold into a building, will be less if it were glazed with PALSUN than for one glazed with glass. This can result in a significant reduction in energy costs both for heating in winter and air-conditioning during the summer. Note that use of Solar Control sheets will insulate just as well, but will also reduce air-conditioning costs because of Near Infra-Red reflection and reduced heat buildup.

Thermal Insulation of PALSUN® vs. Glass

Thickness mm	Thickness in	PALSUN U-Value ($W/m^2 \cdot °K$)	Glass U-Value ($W/m^2 \cdot °K$)
3	(0.12)	5.47	5.81
5	(0.20)	5.19	5.72
6	(0.24)	5.07	5.68
8	(0.31)	4.48	5.60
10	(0.39)	4.63	5.52
12	(0.47)	4.43	5.45

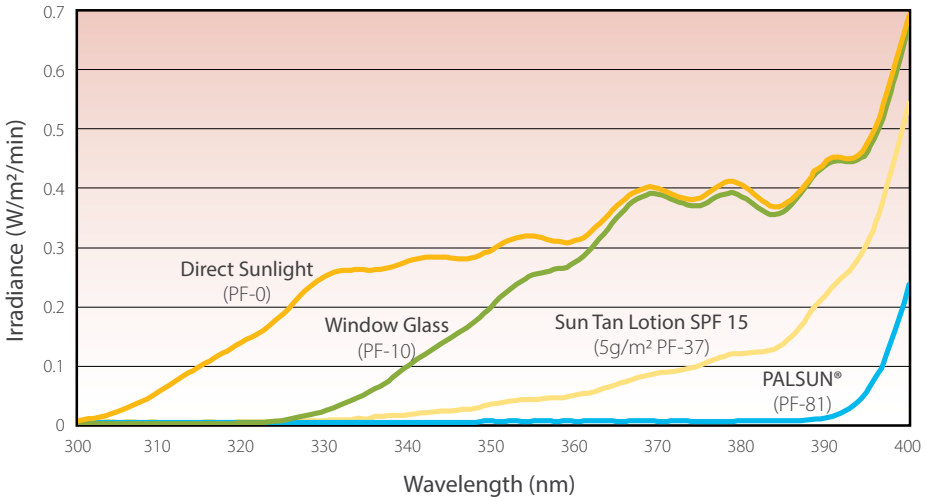
Protection from the Harmful Effects of UV Radiation

Exposure to solar ultraviolet (UV) radiation is widely known as a major health concern by now. The adverse affects were once thought to be associated with solar UV radiation in the 280-315nm (UV-B) range. However, in recent years it has become apparent that exposure to UV-A (315-400nm) is also detrimental. In addition to skin cancer, premature aging has been associated with exposure to UV-A. Both UV-A and UV-B portions of the UV spectrum are blocked out by PALSUN sheets. This screening of harmful UV radiation can be observed in the figure below.

A comparison of the UV protection offered by PALSUN and that offered by sunscreen is indicated in the graph on the right. Note that no barrier is as effective as PALSUN sheet. Activity below PALSUN will be more protected than that offered by proper application of sunscreen, though the latter is sufficient in almost all cases. The key word in the previous sentence is proper. Improperly applied sunscreen or forgetting to apply skin screen will result in undesirable levels of exposure. In addition, note that protection factors are computed on the basis of UV-B exposure. There is as yet no way to compute protection to UV-A exposure. It should also be noted that formulations that only block out UV-B are still being marketed. When playing or swimming below PALSUN, protection is always complete. When swimming, there is no danger that the protection will be washed away. In the last ten years, it has also been documented that UV exposure can also cause damage to the eyes, specifically to the cornea. Wearing sunglasses manufactured from polycarbonate protects the eyes. However, most people remove their glasses when entering the pool. This is a factor for both public and private pools to consider when contemplating a choice of covering.

Irradiance of UV Radiation Trough Various Protective Barriers

(PF=Protective Factor)



Acoustic Properties

Though only about half the weight of an equivalent glass panel, PALSUN glazing offers similar sound insulation properties along with much higher impact strength. These combined properties make PALSUN glazing the preferred material for see-through sound barriers: lightweight, easy to maintain or replace if necessary, highly transparent and vandal-proof.

The table on the right portrays the acoustic performance of PALSUN glazing versus glass.

Acoustic Insulation of PALSUN® Sheets According to EN 1793

Thickness mm	**Sound Reduction** dB
12	31
15	33

Weather Resistance

Solar UV radiation attacks many polymeric materials. The rate of deterioration and crazing on the exterior surface will vary for different polymers. Further erosion is accelerated by water, dirt, air pollution, chemicals etc. The extent of attack depends on environmental factors such as location, altitude, local weather conditions, air pollution etc. The best initial indication is yellowing, followed by a significant reduction in light transmission and structural strength.

All PALSUN sheets (excluding those designated PALTUF, which are UV stabilized) are manufactured with a co-extruded, UV protective layer on one or two sides. This protective layer assures a long lifetime of service. PALSUN sheets retain their toughness and optical quality under intense UV exposure, with minimal reduction in their properties.

2000 hours of accelerated weathering (UV exposure, QUV - ASTM G154) tests, simulating 20 years of exposure in hot, sunny climates cause only a minor decrease in light transmission and a slight increase in yellowness Index for PALSUN. The changes in UV stabilized PALTUF sheet are greater.

The effect of QUV on 3 mm PALSUN & PALTUF sheets appears in the graphs below.

Figure 6a
% Light Transmission Loss of 3mm Sheets

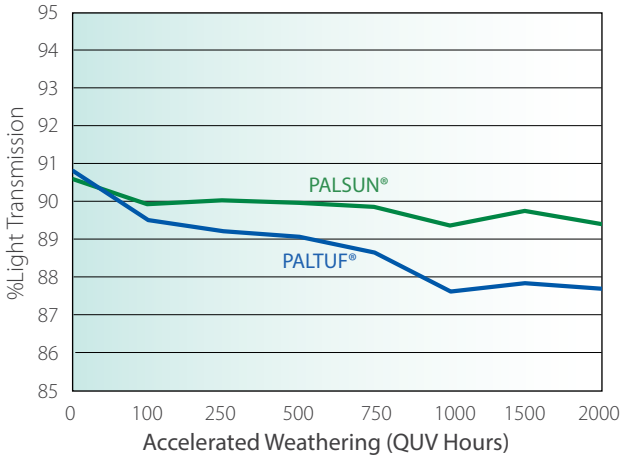
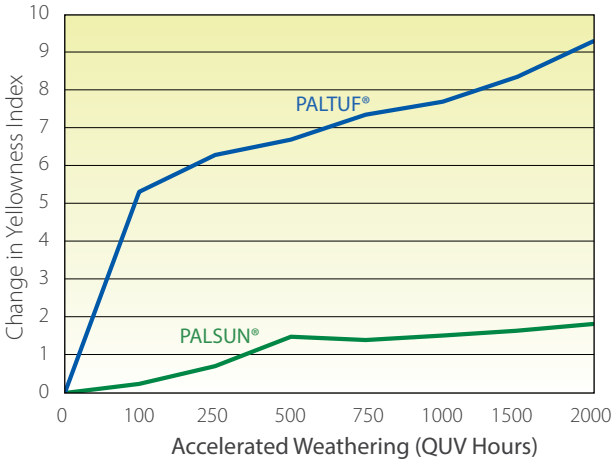


Figure 6b
Yellowness Index Change in 3mm Sheets



Flammability

General

As a thermoplastic, PALSUN eventually melts and burns under the intense heat of a blazing fire. However, PALSUN does not propagate flame, and is solidified and self-extinguished as soon as the direct flame is taken away. PALSUN doesn't produce any toxic fumes or gases when it burns.

PALSUN® FR

PALSUN FR is a fire retardant flat solid polycarbonate sheet with improved flammability ratings. The flame retardant additives make the it virtually non-combustible. When flame licks the sheet, it will only get scorched and eventually melt, solidifying quickly when the direct heat source is removed. Drippings do not ignite other combustible materials, as they do not actually burn.

Smoke and heat extraction

In an actual, full-scale combustion, when PALSUN overhead glazing (as in skylights) is exposed to intense heat it will soften at 150° -160°C and produce apertures in the glazing, enabling heat and smoke to escape. Reduced temperatures inside the structure help to extinguish the fire.

Flammability Classifications

PALSUN and PALSUN FR are classified as appears in the following table, based on tests executed by certified independent testing laboratories.

PALSUN®	
Standard	Classification*
EN 13501	B, s1, d0
NFP 92501, 4	M2 (1.5 to 12mm)
BS 476/7	Class 1y
UL-94	V2 (File e221255)
ASTM D-635	CC1

PALSUN® FR	
Standard	Classification*
UL-94	V-0

* Depends on thickness. For additional information please contact your PALSUN distributor.

General Recommendations for Working With PALSUN®

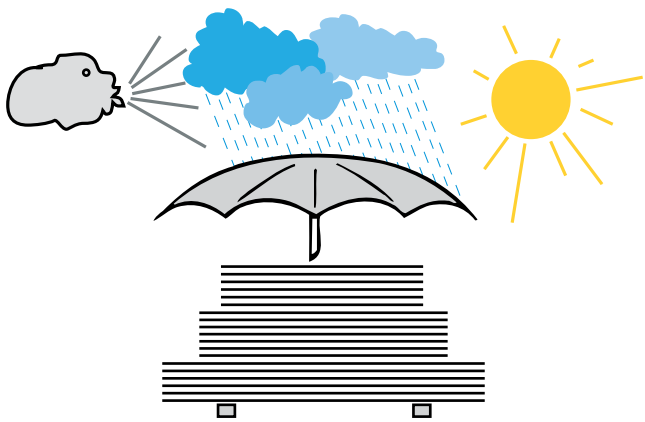
Handling & Storage

1. PALSUN sheets should be transported and stored horizontally, on a flat, sturdy pallet whose dimensions are equal or larger than the largest of the sheets.

   The sheets should be secured to the pallet during transportation and on-site handling. It is possible to stack the sheets with the longer sheets at the bottom and the shorter on top, leaving no unsupported overhang.
2. When moving a pallet with a forklift, always use forks as long as the sheets' width. Shorter forks used on a wider pallet may cause damage to the sheets.
3. PALSUN sheets leave the factory in packages, wrapped in white, watertight polyethylene. The wrapping should be removed as close to the actual time of installation (or use) as possible.

   Storage of the sheets should be in a covered, dry, ventilated place, away from direct sunlight and rain.
4. Avoid extended exposure to direct sunlight, which may cause excessive heat buildup. Long term heating may lead to softening of the protective polyethylene masking, fusing it to the sheet's face and making removal difficult or even impossible.
5. Avoid leaving the sheets stored unwrapped. Dirt may accumulate on the sheets and/or their edges, attracted by electrostatic charges in the sheets, necessitating extra time and labor for cleaning before installation.
6. Whenever necessary to store the pallet in the open, cover it with white opaque polyethylene sheet, cardboard or any other insulating material, taking care to cover the stack completely.

Figure 7: Storing PALSUN® Sheets



Determination of Required Sheet Dimensions

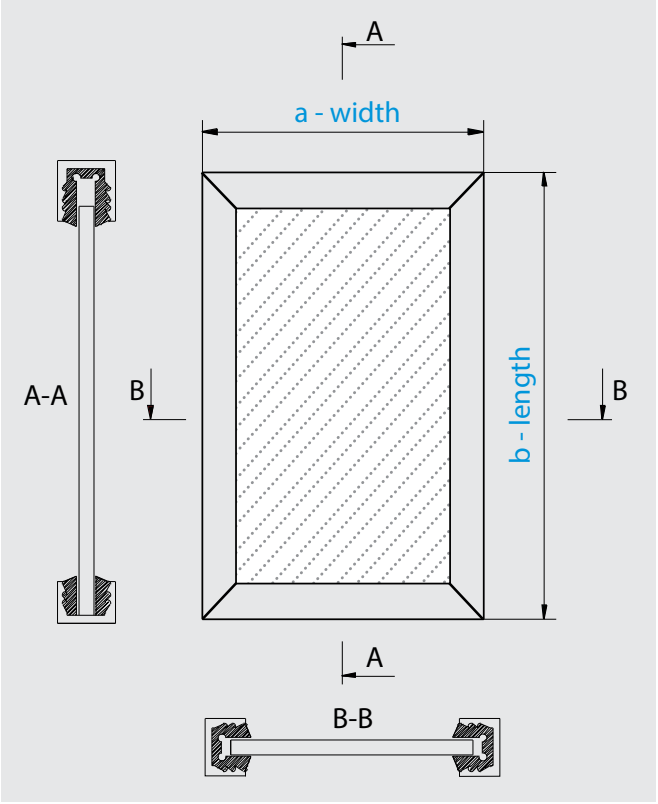
The information below is presented to assist in ordering the required dimensions.

Determination of Sheet Size

Due to thermal expansion, PALSUN & PALTUF sheets have to be cut accurately at predetermined lengths smaller than the dimensions of the frame. At the end of the frame, clearance must be left for expansion. The following tables and diagram help calculate the required sheet dimensions. In addition, there is a table showing the expansion clearance necessary for various sizes of PALSUN & PALTUF sheets.

Determination of Thickness

In order to determine the required thickness, the following table lists the sheet thickness required for a given wind load and width (at the narrow side of the sheet).



PALSUN® Recommended Panel width

Wind/Snow Load Chart for Flat 4 Sides Clamped, Single Panel, Single Span

Thickness	Load (kg/m²)	Length / Width Ratio (b/a)				
		Ratio 1:1	Ratio 1.25:1	Ratio 1.5:1	Ratio 1.75:1	Ratio 2:1
4mm	50	985	875	820	795	785
	75	860	765	715	695	685
	100	780	695	650	630	620
	125	725	645	605	585	575
	150	685	605	570	550	540
	175	650	575	540	525	515
	200	620	550	515	500	490
5mm	50	1235	1090	1025	995	980
	75	1075	955	895	870	855
	100	980	865	815	790	775
	125	910	805	755	730	720
	150	855	755	710	690	680
	175	810	720	675	655	645
	200	775	685	645	625	615
6mm	50	1450	1300	1225	1200	1175
	75	1300	1150	1075	1050	1025
	100	1175	1025	975	950	925
	125	1090	965	910	885	865
	150	1025	910	850	830	815
	175	975	865	810	790	775
	200	925	825	775	755	740

PALSUN® Recommended Panel width - 4 sides clamped (Continued)

Thickness	Load kg/m²	Length / Width Ratio (b/a)				
		Ratio 1:1	Ratio 1.25:1	Ratio 1.5:1	Ratio 1.75:1	Ratio 2:1
8mm	50	1975	1750	1625	1590	1575
	75	1725	1525	1425	1390	1375
	100	1565	1390	1300	1270	1250
	125	1455	1290	1210	1180	1160
	150	1375	1210	1140	1110	1090
	175	1300	1150	1080	1050	1035
	200	1250	1100	1035	1005	990
10mm	50	2050	2050	2050	1990	1960
	75	2050	1910	1795	1740	1715
	100	1960	1735	1630	1580	1555
	125	1820	1610	1515	1465	1445
	150	1710	1515	1425	1380	1360
	175	1625	1440	1355	1310	1290
	200	1555	1375	1295	1255	1235
12mm	50	2050	2050	2050	2050	2050
	75	2050	2050	2050	2050	2050
	100	2050	2050	1955	1895	1870
	125	2050	1935	1815	1760	1735
	150	2050	1820	1710	1655	1630
	175	1950	1730	1625	1575	1550
	200	1865	1655	1555	1505	1480
14mm	50	2050	2050	2050	2050	2050
	75	2050	2050	2050	2050	2050
	100	2050	2050	2050	2050	2050
	125	2050	2050	2050	2050	2025
	150	2050	2050	2000	1930	1900
	175	2050	2000	1890	1835	1810
	200	2050	1925	1810	1755	1730
15mm	50	2050	2050	2050	2050	2050
	75	2050	2050	2050	2050	2050
	100	2050	2050	2050	2050	2050
	125	2050	2050	2050	2050	2050
	150	2050	2050	2050	2050	2025
	175	2050	2050	2025	1950	1925
	200	2050	2050	1925	1875	1850
18mm	50	2050	2050	2050	2050	2050
	75	2050	2050	2050	2050	2050
	100	2050	2050	2050	2050	2050
	125	2050	2050	2050	2050	2050
	150	2050	2050	2050	2050	2050
	175	2050	2050	2050	2050	2050
	200	2050	2050	2050	2050	2050

* For wind load data in N/m² units, multiply value by 10 (e.g. 40 kg/m² = 400 N/m²).

Notes:

1. For 2 sides clamped installation please refer to the ratio 2:1 column.
2. The table is referring to various ratios of Length (b) and width (a) and valid for both Snow and wind load.
3. The table is referring to 4 sides clamped, flat installed Palsun panels.
4. Bedding depth shall be minimum 20mm.
5. The values are calculated according to deflection criterion of L/20 of the short span.
6. The table is not referring to self-weight deflections which might cause aesthetic issues.
7. The table is not referring to the sheet flexibility when vertically installed.
8. The data in the table refer to the short dimension, a.